TiDAR vs AR Throughput (Updated Single-Forward)

Data files loaded dynamically:

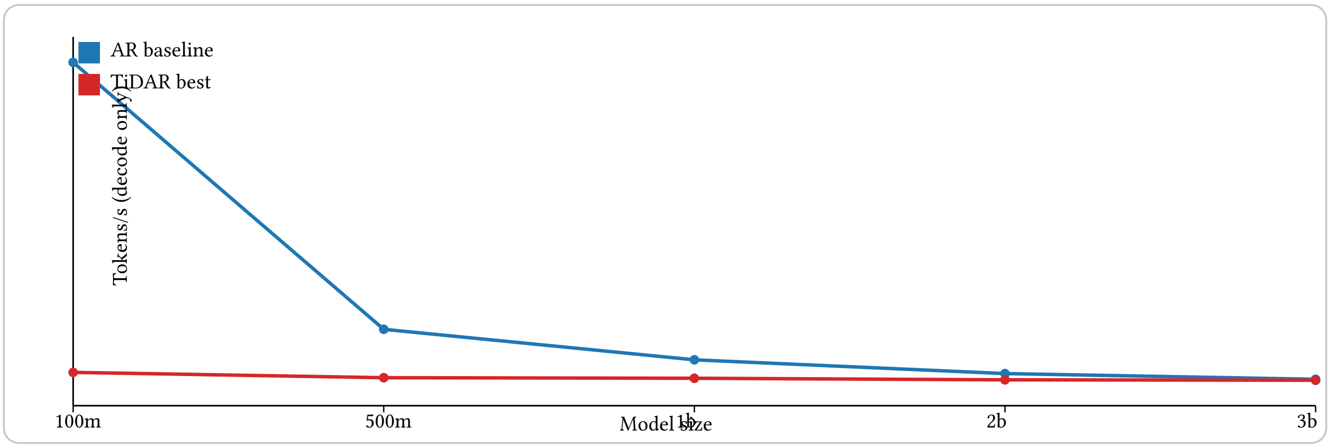
- Benchmark_logs/tidar_vs_ar_single_forward_20260208_134219 + "/best_tidar_vs_ar_by_model_prefill.json"
- Benchmark_logs/tidar_vs_ar_single_forward_20260208_134219 + "/notes.txt"

Sweep:

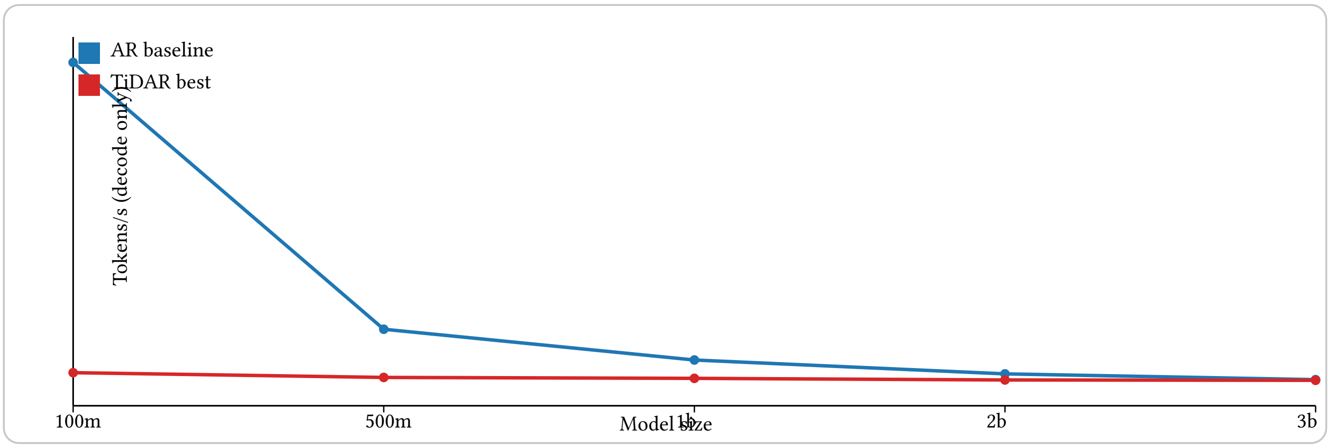
- Models: 100m, 500m, 1b, 2b, 3b
- TiDAR grid: draft_len {4, 8, 16}, accept target {0.6, 0.8}, prefill {0, 1024, 4096}
- Decode steps: 256, repeat: 2
- AR baseline: reused prior run

Line graphs: tokens/s vs model size

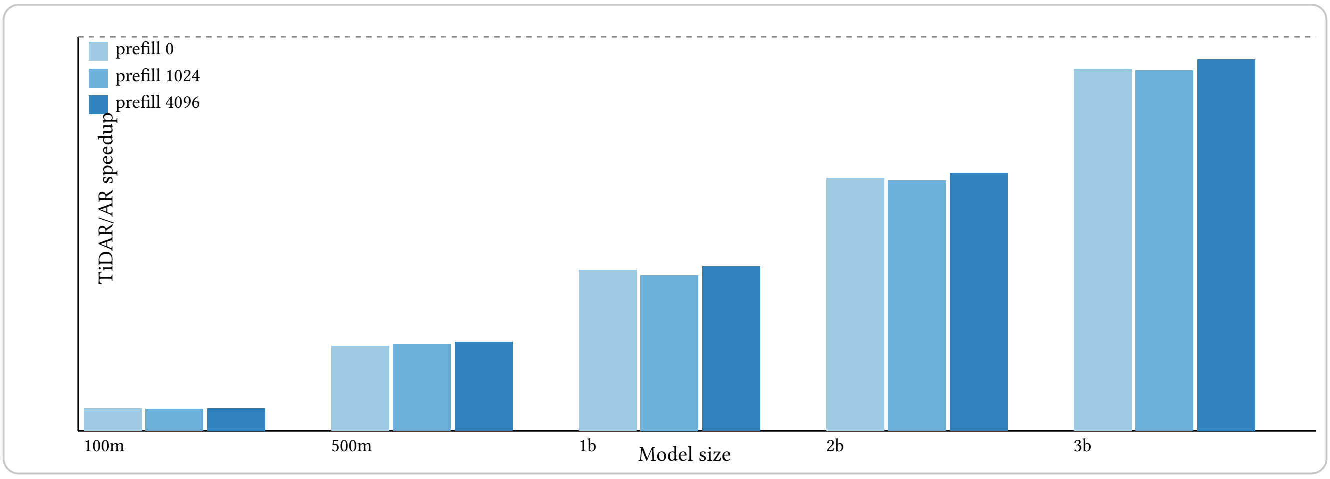
Prefill = 0



Prefill = 4096



Histogram-style grouped bars: TiDAR/AR speedup



Key findings

- Best pair: model 3b, prefill 4096, speedup 0.943x, TiDAR draft 16 at accept 0.8.
- Worst pair: model 100m, prefill 1024, speedup 0.056x.
- Trend in this setup: speedup rises with model size and higher draft, but stays below 1.0x on A40 in current JAX path.

Provenance

Updated TiDAR single-forward sweep vs AR baseline
created_at: 2026-02-08T13:42:19.639340
tidar_summary: TiDAR/Docs/Benchmark_logs/tidar_updated_narrow_20260208_094343/summary.csv
ar_summary: TiDAR/Docs/Benchmark_logs/ar_simple_baseline_sweep_20260207_103142/summary.json
best_global_speedup: 0.9431 at model=3b prefill=4096 draft=16 accept=0.8
worst_global_speedup: 0.0558 at model=100m prefill=1024 draft=4 accept=0.6
pairs_compared: 15

H100 Chat Session Addendum (Mask-Path Comparison)

This section documents the extra H100 PCIe runs executed during this chat so the results are preserved in one place.

What was compared (described by runtime behavior, not git state names):

- Path A — Full-width decode mask path:
  - Decode bias is built over full keys as `[PREFIX | STEP]` (`build_decode_bias_template(cache_len, draft_len, ...)`).
  - Decode attention builds `k_full = [k_prefix, k_step]`.
  - Per-step prefix validity is added as explicit bias, then one dense `dot_product_attention` call is run.
  - This is the current baseline inference path used in the earlier H100 sweep.
- Path B — Local experimental mask-layout branch:
  - Decode bias source is switched to step-first layout (`build_decode_bias_template_step_prefix(...)`) in inference/throughput code.
  - Attention decode path in `Transformer_block.py` is expanded with:
    - single-call step-first K/V path using `key_value_seq_lengths`,
    - legacy fallback for prefix-first full-width masks,
    - step-only split-attention merge path (prefix attention + step attention).
  - Additional step-only bias builders were added in `tidar_core.py` and prefill bias wiring was changed to step-only for prompt+draft prefill.

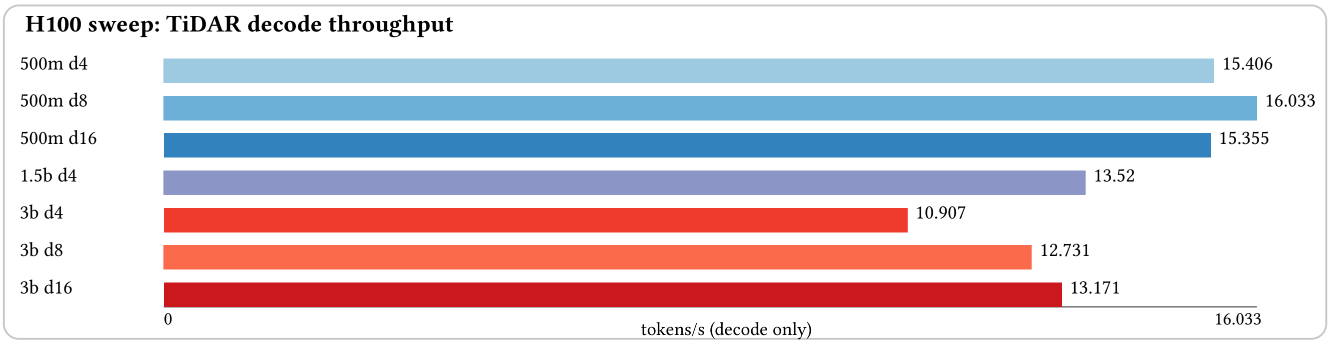
Hardware and run settings

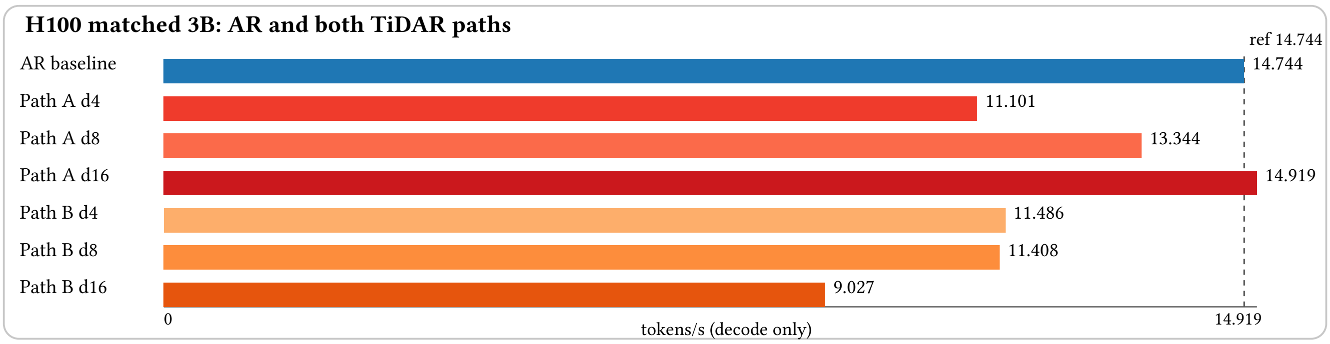
- GPU: NVIDIA H100 PCIe (81,559 MiB)
- Remote host alias used: `gpu-box-1`
- Decode context setting: `context_length = 4608`
- Prefill setting: `prefill_tokens = 4096`
- Decode steps: `256`
- Seed: `0`
- Acceptance target for TiDAR throughput tests: `accept-rate = 0.8`

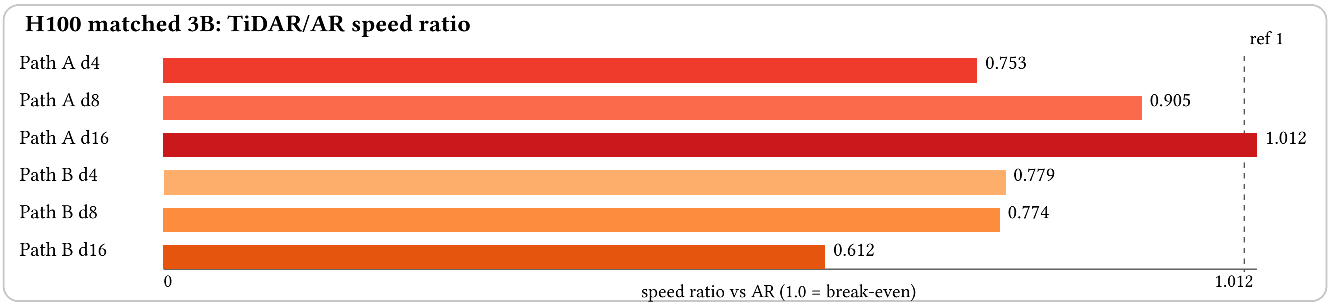
Scripts used in this chat

- TiDAR throughput script:
  - `TiDAR/model/test_inference_throughtput.py`
- AR decode-only helper used for matched 3B comparison:
  - `/tmp/ar_decode_throughput_current.py`
  - (temporary helper; uses the same TiDAR model stack and KV-cache path for one-token AR decode)

Visual summaries







Recorded H100 metrics from this chat

First H100 TiDAR sweep (Path A, decode-only tokens/s):

- 500m, draft 4: `15.405662` tok/s
- 500m, draft 8: `16.033103` tok/s
- 500m, draft 16: `15.354966` tok/s
- 1.5b, draft 4: `13.519934` tok/s
- 3b, draft 4: `10.906801` tok/s
- 3b, draft 8: `12.730544` tok/s
- 3b, draft 16: `13.171004` tok/s

Matched 3B run (same settings above), AR + TiDAR path comparison:

- AR baseline (decode-only):
  - generated tokens: `256`
  - iterations: `256`
  - decode time: `17.362658 s`
  - throughput: `14.744286 tok/s`
- TiDAR, Path A, draft 4:
  - generated tokens: `256`
  - iterations: `78`
  - observed avg accept/iter: `3.282051`
  - observed max accept/iter: `4`
  - decode time: `23.061257 s`
  - throughput: `11.100869 tok/s`
- TiDAR, Path A, draft 8:
  - generated tokens: `256`
  - iterations: `39`
  - observed avg accept/iter: `6.564103`
  - observed max accept/iter: `8`
  - decode time: `19.184134 s`
  - throughput: `13.344361 tok/s`
- TiDAR, Path A, draft 16:
  - generated tokens: `256`
  - iterations: `21`
  - observed avg accept/iter: `12.190476`
  - observed max accept/iter: `16`
  - decode time: `17.159882 s`
  - throughput: `14.918518 tok/s`
- TiDAR, Path B, draft 4:
  - generated tokens: `256`
  - iterations: `78`
  - observed avg accept/iter: `3.282051`
  - observed max accept/iter: `4`
  - decode time: `22.288248 s`
  - throughput: `11.485874 tok/s`
- TiDAR, Path B, draft 8:
  - generated tokens: `256`

  - iterations: `39`
  - observed avg accept/iter: `6.564103`
  - observed max accept/iter: `8`
  - decode time: `22.439773 s`
  - throughput: `11.408315 tok/s`

- TiDAR, Path B, draft 16:
  - generated tokens: `256`
  - iterations: `21`
  - observed avg accept/iter: `12.190476`
  - observed max accept/iter: `16`
  - decode time: `28.360653 s`
  - throughput: `9.026591 tok/s`

Quick interpretation from these runs

- In this matched 3B run, Path A at draft 16 slightly exceeded AR baseline throughput (`14.918518` vs `14.744286` tok/s).
- Path B improved only draft 4 slightly, but regressed strongly for draft 8/16.
- Net result from this chat: the current local experimental branch did not improve overall TiDAR throughput on H100 in the tested settings.